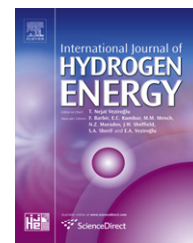


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Experimental and comparative study of metal hydride hydrogen tanks

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ABSTRACT

Hydrogen, as an energy carrier, is a solution seriously considered at worldwide scale to enable renewable energy to be a part of our quotidian. Several challenges remain to produce, to store and to use hydrogen effectively. In our laboratory, we are interested in the discipline of hydrogen storage which is the subject of this paper. Two hydrogen storage tanks based on metal hydrides were built at the laboratory respecting a simple and modular design in a purpose of getting useful knowledge about the similarity elements of metal hydride tanks relevant in the prediction of a larger size storage unit behaviour. A description of the tanks is presented and an investigation of the parameters influencing the hydrogen storage performance is done.

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1. Introduction

Hydrogen absorbing alloys can be applied for industrial applications that have considerable potential, such as for hydrogen storage, chemical compressors and chemical heat pumps. This is because it has features like: generating a large amount of heat with hydrogen absorbing, absorbing and desorbing hydrogen reversibly and storing hydrogen compactly [1].

The dynamic response of hydrogen storage system is important in the design of metal hydride energy conversion systems. However, it is known that the performance of hydrogen absorption to and desorption from metal hydride depends on the heat transfer, mass transfer and kinetic-reactions of the metal hydride bed. So it is necessary for the design of optimized MHV to predict transient heat and mass transfer in the metal hydride bed. Hence, there have been numerous experimental and theoretical investigations on

several aspects of hydride storage. Gopal et al. [2,3,24] and Mayer et al. [4] investigated transient heat and mass transfer within the reactor using a one-dimensional model. Jemni and Ben Nasrallah [5,6] examined both the dimensional heat and mass transfer within a metal-hydrogen reactor for both absorption and desorption processes using two mathematical models. The sorption process for mixtures containing impurities that are inactive or weakly active to the hydride forming material was investigated by Shmalkov et al. [7]. Mat et al. [8] and Aldas et al. [9] studied numerically three-dimensional heat and mass transfer in a metal hydride bed during absorption. Jemni et al. [10] conducted an experimental and numerical study in order to determine the effective thermal conductivity, the equilibrium pressure, and the kinetic-reactions. Demircan et al. [11] examined experimentally the hydrogen absorption in two $\text{LaNi}_5\text{-H}_2$ reactors. Kikkinides, E.S et al. [12] presented a comprehensive review is performed on all possible design configurations and heat management

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strategies in metal hydride storage tanks. The main conclusion is that an enhanced cooling during hydrogen filling of the storage tank improves significantly hydrogen storage time requirements. Mellouli et al. [13] designed and manufactured a novel design of MHV equipped with a spiral heat exchanger and mentioned that the charge and the discharge time are significantly reduced. Recently, S. Kikkinides, E.S et al. [14], developed effective heat management strategies and novel cooling design options for hydrogen storage systems in LaNi_5 metal hydride reactors by performing systematic simulation and optimization studies. In a previous publication [15], we have measured and modelled the kinetics of hydrogen sorption by LaNi_5 and two related pseudobinary compounds. M. Melnichuk et al. [16] optimized the aluminium fraction of heat transfer fins for a LaNi_5 hydrogen storage container, at different absorption times and container diameters. Askri et al. [17], established a model to study the dynamic behaviour inside various designs of MHVs and optimization results indicate that almost 80% improvement of the storage time can be achieved when the design includes a concentric heat exchanger tube equipped with fins and filled with flowing cooling fluid.

All these studies have shown that the charge as well as discharge times of MHV strongly depend on the successful heat transfer from/to the hydride bed. Thus, different attempts to improve the effective thermal conductivity of metal hydride beds have been undertaken: insertion of aluminium foam [18], integration of copper wire net structure [19], packing in a multilayer waved sheet structure, micro-encapsulated metal hydride compact [20], and compacting metal hydride powder with an expanded graphite [21].

In addition, the effect of different parameters (flow mass and temperature of the cooling fluid, applied pressure, and hydrogen container volume) has been discussed and the outcome shows that a good choice of these parameters is vital.

A. Chaise et al. [22] have improved the volumetric hydrogen storage capacity of magnesium hydride by compacting it into disc shape and they have improved the thermal conductivity by adding expanded natural graphite (ENG). The loading time has been considerably reduced by adding 20% ENG to the compacted disks; furthermore, the weight capacity is markedly improved.

C. Panos et al. [23] presented a systematic framework for the optimal design, optimization and advanced control of the desorption process in metal hydride bed reactors. The framework features: (i) a detailed two-dimension dynamic process model, (ii) a design and operational dynamic optimization step, and (iii) an explicit/multi-parametric model predictive controller design step. For the controller design, a reduced order approximate model is obtained, based on which nominal and robust multi-parametric controllers are designed.

2. Experimental procedure

2.1. Metal hydride containers design

Both designed tanks, T_1 and T_2 , have a cylindrical geometry which represents the envelope of the hydride bed.

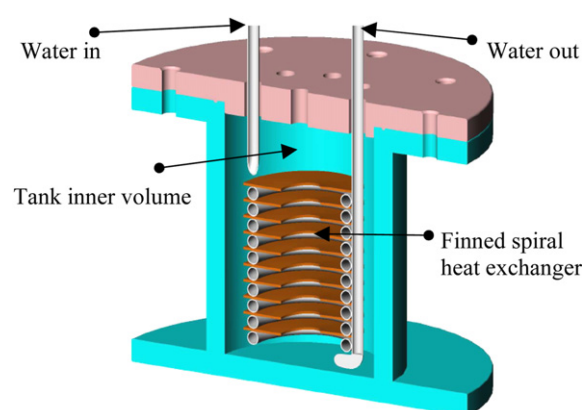


Fig. 1 – Cross section of T_1 tank.

The T_1 tank has an inner volume of 110 mm in height and 80 mm in diameter (Fig. 1). A finned spiral heat exchanger is immersed in the T_1 hydride bed. This thermal exchanger consists of a stainless steel tube of inner and outer diameters of 5 and 6 mm respectively, which is rolled up into a circular helix with 50 mm of diameter and 11 turns with a pitch of 7 mm. Thus, taking into account the outer diameter of the tube and the value of the pitch, we obtain a space between two turns of 1 mm in which we inserted a copper fin coil.

About the T_2 tank (Fig. 2), the inner volume is 120 mm in height and 60 mm in diameter. A tube equipped with eight circular fins, all made of stainless steel, is mounted coaxially inside the T_2 tank to provide heat exchange between the hydride powder that surrounds it and water coolant flowing through it.

Both tanks have an O-ring sealed flange at one end which has a hydrogen outlet/inlet orifice fitted with a porous metal filter. Also, they are well insulated (insulation is not represented). The dimensional of the filter are: Diameter: 10 mm, Thickness: 2 mm, Porosity: 20 μm .

2.2. Experimental set-up

The experimental set-up is a Sievert-type experimental apparatus mainly consists of a hydride reactor, two reference cylinders interconnected via valves, a vacuum pump,

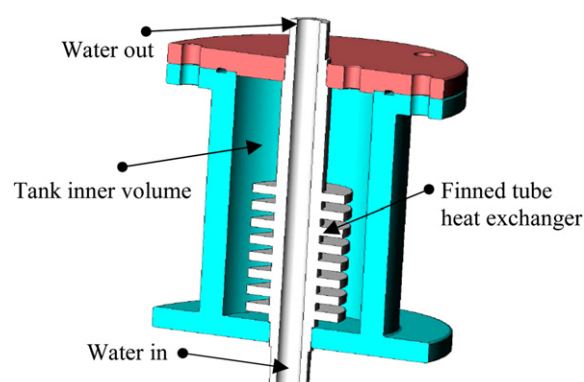


Fig. 2 – Cross section of T_2 tank.

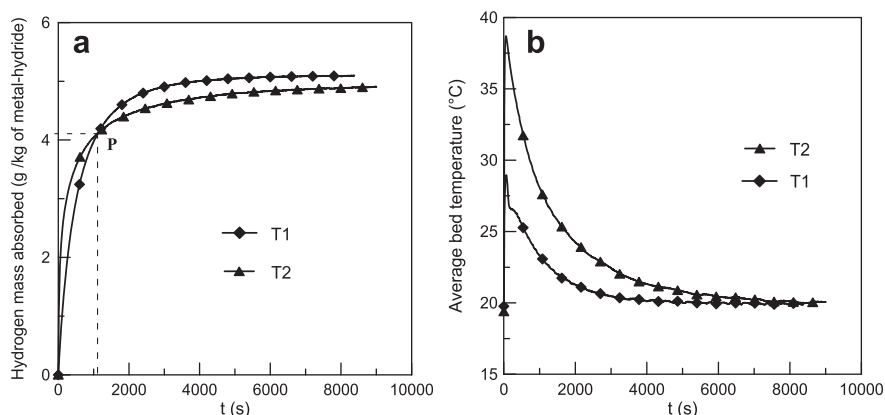


Fig. 3 – Time history of (a) the absorbed hydrogen mass and (b) the hydride bed temperature for both tanks T_1 and T_2 .

a thermostatic water bath (0–70 °C), a water pump, a data acquisition system and a 99.99% pure hydrogen gas cylinder. The temperature and pressure acquisition are recorded on a computer for further processing and interpretation.

2.3. Conduced experiments

The experimental procedure for the absorption begins by starting the vacuum pump to maintain a low pressure in the reactor (0.01 bar) at high temperature (50 °C) for a long period of time (4 h) to ensure that the sample of LaNi_5 is emptied of hydrogen (or other gases). Then, thermostated water flow through the heat exchanger to bring the tank to the temperature value which will be the initial and final state of absorption test. The reference volume is fixed and then loaded with hydrogen at a desired pressure. The tank and the reference volume are then interconnected and the data acquisition (hydrogen dynamic pressure and temperatures of hydride bed and water outlet) is executed.

3. Experimental results

In the following discussion, the storage performance of a tank denotes mainly the hydrogen storage capacity and the absorption rate.

3.1. Effect of heat exchanger design

Fig. 3 shows the time history of the stored hydrogen mass in tanks T_1 and T_2 and the average hydride beds temperature of both tanks.

As illustrated by Fig. 3a, the hydrogen storage performances of both tanks are comparable. Furthermore, there is a point P at which both tanks store the same amount of hydrogen at exactly the same period of time. To the left of P point, the T_2 tank has a storage performance relatively better than the T_1 tank while on the right of point P, the T_1 tank takes over.

For hydrogen storage tanks based on metal hydrides, the driving force for mass transfer is identified as the difference between the supply hydrogen pressure P_{app} and the equilibrium pressure P_{eq} which depends mainly on the type of hydride and its temperature. If the hydride is heated, P_{eq} increases, hence the driving force for mass transfer decreases, i.e., the hydrogen storage performance deteriorates and vice versa if the hydride is cooled [24].

Fig. 3b reveals that the temperature of the T_1 hydride bed is permanently lower than that of the T_2 hydride bed. So, the finned spiral heat exchanger is more efficient than the finned tube heat exchanger in removing the heat of absorption. This can explain the behaviours of both tanks but only after reaching the P point: the better storage performance of T_1 tank is due to the higher heat transfer rate of the finned spiral heat

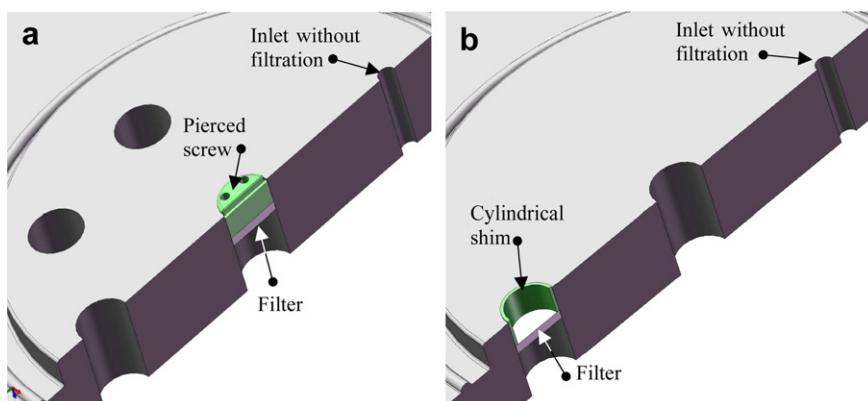


Fig. 4 – Cross sections of (a) T_1 lid and (b) T_2 lid.

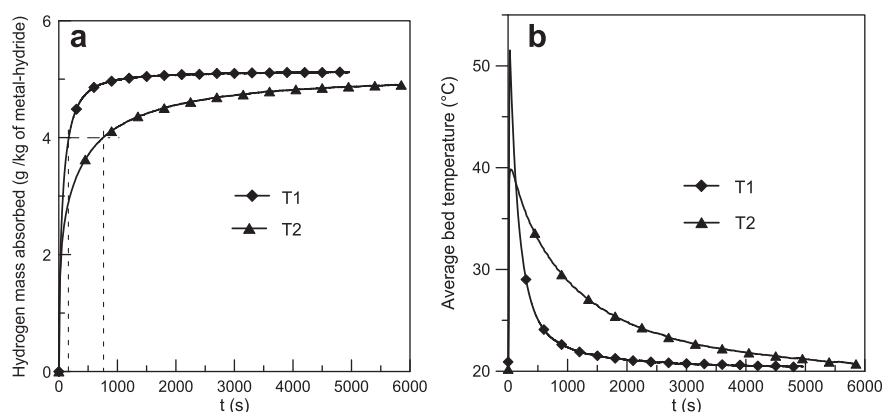


Fig. 5 – Time history of (a) the absorbed hydrogen mass and (b) the hydride bed temperature for both tanks T_1 and T_2 .

exchanger, reducing the equilibrium pressure of the hydride bed and enhancing the driving force for mass transfer [25]. On the other side, before reaching the point P, we can't rely exclusively on the data of the hydride beds temperatures to compare between the two tanks and to analysis their behaviours. This conclusion leads us to investigate the influence of other elements of a tank on its storage performance.

3.2. Effect of the inlet hydrogen design

During the storage process, the tank inlet orifice is the first element in contact with the flow of hydrogen. For T_1 tank, hydrogen is introduced through a cylindrical orifice with a sintered steel filter held in position by a screw whose head is pierced with 4 holes (Fig. 4a). The T_2 tank has also a cylindrical orifice with a sintered steel filter held in position by a cylindrical shim and two screws (Fig. 4b).

To assess the influence of each inlet design on the storage performance, another cylindrical orifice without filtration was added to each tank. We repeat the experiment of Fig. 3 using the new inlet design for introducing hydrogen in both tanks (Fig. 5).

By comparing between Figs. 3 and 5a, immediately, we note the significant improvement of storage performance of T_1 tank. For example, the storage duration of 4g of hydrogen goes down from 1000 to 200 s. Thus, we conclude that the hydrogen inlet design of T_1 tank reduces the absorption rate. For the T_2 tank, the improvement is less significant, since the storage time of 4g of hydrogen goes down from 1000 s to 800 s. Thus, we conclude that the hydrogen inlet design of T_2 tank is compatible with the others tank elements (heat exchanger, etc.), since it does not inhibit the storage performance.

In fact, the task of designing and dimensioning the hydrogen inlet of a tank should be seriously taken into consideration because it can induce a resistance to the hydrogen flow and reduces the hydrogen pressure inside the tank, thus, the driving force for mass transfer decreases. Actually, a filter is required in the design of the hydrogen outlet of a metal hydride tank since it prevents hydride to leave the confines of the tank during a desorption process.

About the impact of the hydrogen inlet design on the temperature evolution of both hydride beds, note the high variation of the temperature peak value of T_1 hydride bed

(Figs. 3 and 5b). This can be considered as an indicator of the improvement of the hydrogen flow entering T_1 tank, since the peak temperature rise reflects an increase in absorption heat which means that more hydrogen was absorbed.

From Fig. 5a, it is clear that the T_1 tank is more efficient in hydrogen storing than the T_2 tank, and this throughout the experiment length. In this case, the behaviours of the two tanks are mainly resulting of the difference between the efficiencies of the two heat exchangers in cooling the respective hydride beds.

3.3. Effect of coolant temperature and hydrogen supply pressure case of T_2 tank

Lowering the cooling temperature (Fig. 6) helps to cool more effectively the sample of LaNi_5 , therefore, accelerate the kinetics of the exothermic absorption reaction. In fact, if we decrease the temperature of the hydride metal, the equilibrium pressure decreases too, therefore the driving force for mass transfer increases.

By setting the coolant temperature (20 °C) and water flow (13 g/s), we can evaluate the effect of the variable pressure on the absorption phenomenon (Fig. 7). Increasing the initial

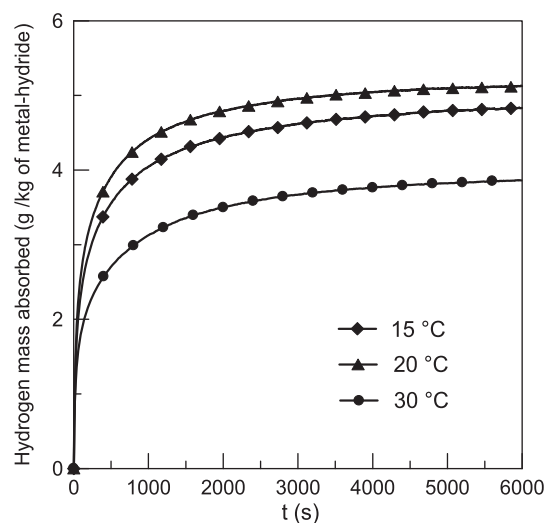


Fig. 6 – Effect of the cooling water temperature pressure on the storage performance.

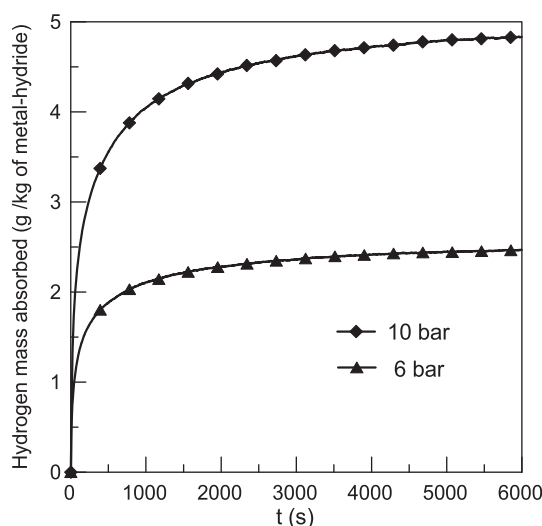


Fig. 7 – Effect of the initial hydrogen supply pressure on the storage performance.

hydrogen pressure at constant reference volume has the effect of improving the absorption kinetic and the mass of absorbed hydrogen. In this case, the supply pressure is the responsible for raising the driving force for mass transfer since the equilibrium pressure is set by the initial experimental conditions.

4. Conclusion

The experimental study above showed that the hydrogen storage in metal hydride tanks is mainly heat transfer dependent, thus, the tank with better cooling system exhibits the fastest storing kinetics. The conducted measurements revealed that an appropriate hydrogen inlet design improves, also, the hydrogen storage performance. In addition, the hydrogen storage can be improved by a proper choice of the operating temperature and the supply pressure.

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